

Mathematics for Machine Learning

Vazgen Mikayelyan



Limit of a Function

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Definition

Let $f : X \rightarrow \mathbb{R}$, $X \subset \mathbb{R}$ is an interval and $a \in X$. It is said the limit of f , as x approaches a , is A and written $\lim_{x \rightarrow a} f(x) = A$ or $f(x) \xrightarrow{x \rightarrow a} A$, if for all $\varepsilon > 0$ there exists $\delta > 0$ such that from $0 < |x - a| < \delta$, $x \in X$ follows that $|f(x) - A| < \varepsilon$.

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- 1 What if $X = (0, 1]$ and $a = 0$?
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Continuous Functions

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- 1 What if $X = (0, 1]$ and $a = 0$?
- 2 What about properties of continuous function?

Theorem

Let $f \in C[a, b]$. If $f(a)f(b) < 0$ then there exists $c \in (a, b)$ such that $f(c) = 0$.

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If $f \in C[a, b]$, then f has maximum and minimum values.

Uniform Continuity

Definition

Let $f : X \rightarrow \mathbb{R}$, $X \subset \mathbb{R}$. It is said f is uniformly continuous on the set X , if for all $\varepsilon > 0$ there exists $\delta > 0$ such that from $|x_1 - x_2| < \delta$, $x_1, x_2 \in X$ follows that $|f(x_1) - f(x_2)| < \varepsilon$.

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Theorem

If $f \in C[a, b]$, then it is uniformly continuous on $[a, b]$.

Definition

Let $f : X \rightarrow \mathbb{R}$, $X \subset \mathbb{R}$. It is said f is differentiable at interior point $x_0 \in X$, if the following limit exists

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Proposition

$$f(x_0 + h) - f(x_0) = f'(x_0)h + o(h), h \rightarrow 0.$$

Derivative

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Theorem

If f has a finite derivative at x_0 then it is continuous at x_0 .